

The Impact of Context and Choice on the Boundary-Spanning Process: An Empirical Extension¹

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This paper develops and tests a theoretical model of the relationships among environmental interdependence, environmental uncertainty, and boundary-spanning activities. Drawing from the strategic choice literature, a decision variable labeled influence on organizational decision was incorporated as a moderator. Results supported the environmental interdependence–environmental uncertainty–boundary-spanning activities relationships, but the individual influence variable failed to emerge as a moderator.

INTRODUCTION

Interdependence has been discussed by several modern management theoreticians including Pfeffer & Salancik (1978), Straw and Szwajkowski (1975), and Thompson (1967). March and Simon (1958) originally included the effect of environmental munificence, an aspect of dependence, on intra-organizational conflict. Several other authors, (e.g., Straw & Szwajkowski, 1975; Cyert & March, 1963; Osborn & Hunt, 1974; Osborn, 1976) postulate theories on the impact of dependence on a variety of organizational variables but seldom provide empirical support for their theoretical frameworks.

¹The program leading to this article was supported in part by a grant from University of Petroleum and Minerals.

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Major empirical studies have viewed organization environments primarily in terms of uncertainty (e.g., Lawrence & Lorsch, 1969; Duncan, 1972). This research appears to deny what several notable authors (Thomson, 1967; Pfeffer, 1972) have proposed regarding environmental uncertainty. These scholars suggest that environmental uncertainty is actually the result of perceived dependence on environmental elements. Pfeffer and Salancik (1978) posit that dependence is causally related to environmental uncertainty. This study attempts to empirically validate Pfeffer and Salancik's contention by empirically investigating the relationships among generally acknowledged dimensions of dependence (substitutability and essentiality, Jacobs, 1974) and environmental uncertainty (dynamisms and complexity, Lawrence & Lorsch, 1969; Duncan, 1972).

Knowledge of the environment is only useful if it results in improved organizational activities. According to Thompson (1967), the primary conveyers of environmental input to organizational decision makers are the boundary spanners who operate at the environment-organization interface. This suggests that the immediate environmental impact is on the activities of individuals at the organizational boundary.

Boundary spanners are individuals employed in boundary units. To the extent that these individuals can exercise discretion at contingent boundaries that are important to organizational survival, the individual has influence over organizational activities. Prior empirical research supports the influence of key individuals on organizational decision making (Reimann, 1975; Schermerhorn, 1975; Tushman & Scanlon, 1981a, b; Schwab, Ungson, & Brown, 1985). However, a dearth of research exists that relates the individual to the environment-boundary-spanning activities relationship. Various researchers such as Strauss (1962), Keller, Szilagyi, and Holland (1976), Schwab et al. (1985), Miles (1976), and Cox, Hitt, & Stanton et al. (1978) have dealt with aspects of boundary spanning. No prior boundary-spanning studies (i.e., Strauss, 1963; Keller et al., 1976; Schwab et al., 1985; Miles, 1976; Cox et al., 1978) recognize the individual characteristics of the boundary spanner as a possible moderator of boundary-spanning activities.

The model illustrated in Fig. 1 posits that environmental dependence is antecedent to environmental uncertainty (Pfeffer & Salancik, 1978) and that both operate to determine the boundary-spanning activities engaged in by boundary agents. In addition, specific individual characteristics of these agents moderate how the boundary spanner relates perceptions of dependence to uncertainty and uncertainty to boundary activities.

Emerson (1962) used two concepts to define interdependence. These were *A*'s motivational investment in goals mediated by *B* and the availability of these goals outside the *A-B* relationship. Jacobs (1974) translated Emerson's concepts into two aspects of interdependence; substitutability and essentiality. The former is defined as the degree to which an environmental

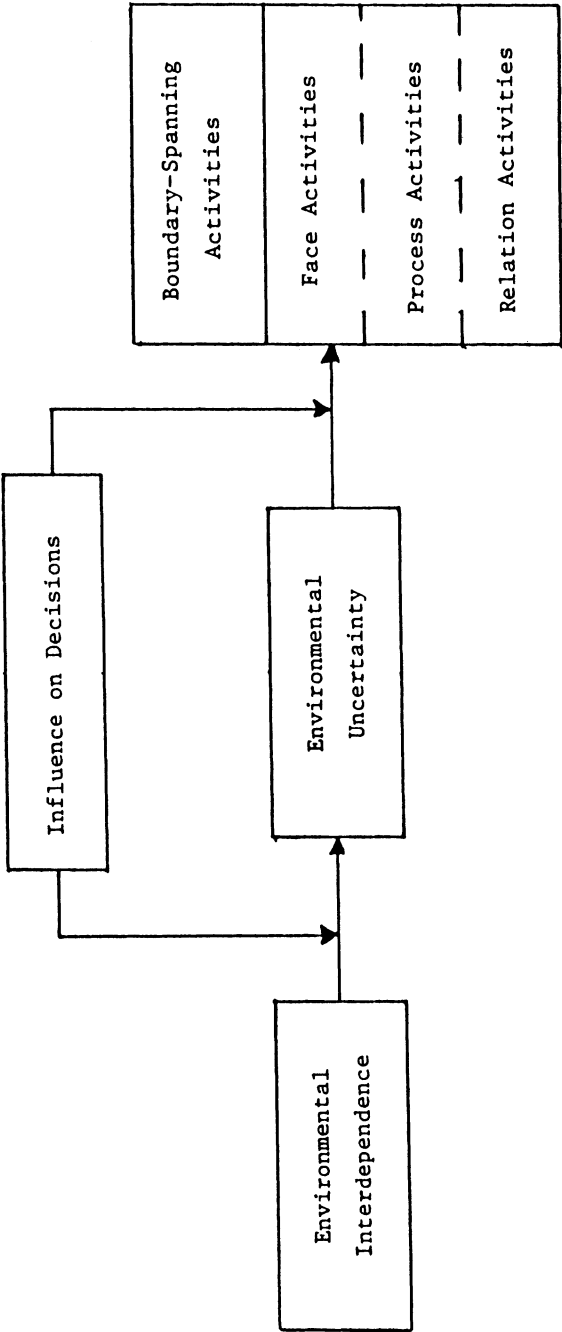


Fig. 1. Conceptual model: Relationship between individual characteristics, interdependence, environmental uncertainty, and boundary-spanning activities.

element could be replaced by another element. The latter is defined as the degree to which an element is essential to the organization's functions. Jacob's refinement of Emerson's construct will be used to operationalize interdependence in this study.

Environmental uncertainty has been described in terms of complexity and dynamism (Thompson, 1967; Lawrence & Lorsch, 1969; Duncan, 1972). Few studies investigated the relationship between uncertainty and boundary spanning (Leifer, 1975; Leifer & Huber, 1977) which is the focus of the research reported here. Boundary spanning is conceptualized using Jemison's (1979) list of 25 Boundary-Spanning Activities (BSA). There are three, *a priori*, activity factors delineated within the list of 25 activities: face, process, and relation. Face activities relate to representing or protecting the focal organization. Activities that involve environmental scanning, monitoring, or gatekeeping are process activities. Last, relation activities relate to the transacting, linking, and coordinating activities of the boundary spanner. Jemison argues that this list represents the major factors and activities associated with boundary spanning.

The model presented in Fig. 1 depicts an individual characteristic, influence on decisions (IOD), as a moderator of both causal relationships. Thompson (1967) argued that boundary spanners who control important segments of the task environment create an organizational dependence upon them by which they acquire more influence in organizational decision making. The conceptualizations of environmental interdependence and boundary spanning developed by Jacobs (1974), Emerson (1962), and Jemison (1978) appear to support Thompson's arguments.

In brief, it is argued that a positive significant relationship exists among interdependence (substitutability and essentiality), environmental uncertainty (dynamism and complexity), and boundary-spanning activities (face, process and relation). Furthermore, boundary spanners' influence on decisions moderates both relationships specified in the model. The following sections of this paper report the method and results of an empirical investigation of the model developed above. The objectives for this study are: (1) to develop a theoretical framework that incorporates the notions of environmental interdependence and uncertainty as posited by Pfeffer and Salancik (1978) with the moderating influence of the boundary spanner to determine his/her boundary activities, and (2) to conduct an empirical investigation to test the relationships specified in the theoretical framework.

RESEARCH METHODS

The contextual factors related to boundary spanning have been extensively investigated (Jemison, 1978; Miles, 1980; Schwab et al., 1985) via sever-

al correlational studies in both field and laboratory settings (Cox et al., 1978; Leifer, 1975). General agreement exists on the nature of relationships, e.g., environmental uncertainty and boundary-spanning activity (Cox et al., 1978), and influence on decisions and boundary spanning (Jemison, 1978); however, the direction of these relationships remains untested. Therefore, this field study investigated the unidirectional effect of environmental interdependence, environmental uncertainty, and boundary-spanning activity. To test for the moderating effects of influence on decisions, the relationships were analyzed using moderated regression analysis (Peters & Champoux, 1979; Zedeck, 1971).

The sample consisted of 133 (41% usable response rate) purchasing agents from electronics firms in the southwest United States. The electronics industry was selected because it is dynamic and displays a high level of intensity of environmental interaction. Purchasing agents were chosen because they assume a boundary-spanning role. Thus, their activities entail all of the 25 boundary-spanning activities in Jemison's (1979) conceptualization.

To operationalize interdependence as conceptualized by Jacobs (1974), a 12-item index was developed based on previous literature (Duncan, 1972; Al-Mubarak, 1982). This index tapped the substitutability and essentiality of the environment using 5-point Likert scales. Given that the environmental interdependency scale was developed for this study, it was necessary to verify that the items generated actually tapped the underlying dimensions of the interdependence construct. To achieve the above objective, an exploratory factor analysis with varimax rotation was used. This analysis was intended to substantiate the construct validity of the scale for this sample of boundary spanners (purchasing agents).

Table I presents the results of the factor analyses for the interdependence scale. A conservative loading of approximately 0.50 was used to determine if an item should be included in a factor. The design of this scale utilized a variation of the ecomethodology introduced by Miles (1980). This scale construction methodology alters the use and interpretation of factor analysis for construct validation. That is, rather than confirm the factor structure of *a priori*, ubiquitous scale items, factor analysis can be used in an exploratory manner to verify the construct validity of the items selected for that unique sample.

In the case of the study reported here, the sample consisted of 133 purchasing agents from electronics firms in the United States (41% usable response rate). *A priori*, one would expect that they would consider their function most dependent on the input or supplier segment of the environment. As Table I indicates, all four supplier-related essentiality items loaded unambiguously on Factor 1, thus confirming that "this sample" considered suppliers most essential or important to the success of their firm. A similar conclusion can be drawn regarding Factor 2. Three of the four supplier types

Table I. Principal Factor Analysis with Varimax Rotation—Final Interdependence Scale

Item	Factor			Communality
	1	2	3	
Essentiality				
New materials suppliers	.705	.065	.055	.174
Equipment suppliers	.800	.034	.221	.517
Parts suppliers	.653	.310	.156	.510
Labor supply	.494	.188	.075	.384
Substitutability				
Equipment suppliers	.219	.519	.464	.447
Parts suppliers	.120	.627	.057	.280
Labor supply	.078	.658	.232	.612
Competitors for customers	.133	.550	.323	.456
Eigenvalues	3.33	1.53	.87	
Percent variance explained	52.2%	24.0%	13.6%	
Cumulative variance	52.2%	76.2%	89.8%	

showed high substitutability while the “competition for customers” item reflected the nature of the electronics firms surveyed. That is, the firms included in this study were large suppliers of semiconductors to larger computer manufacturers and/or the federal government. Thus, the relative lack of substitutability or high interdependence on one or a few customers in a highly competitive industry was apparent to all employees. Thus, the factor analysis results in Table I appear to verify the ecological construct validity of this scale in this environmental context.

In a further attempt to assess the appropriateness of the newly developed interdependence scale, internal reliabilities were calculated for both the substitutability and essentiality subscales. Reliabilities achieved were .80 (Cronbach alpha) for both subscales. These values exceed the criterion of 0.70 established by Nunnally (1978) for exploratory scale development.

Previously published and generally accepted scales were used where available for other constructs included in this study. The two dimensions of environmental uncertainty (dynamism and complexity) were measured using Duncan's (1972) scale. Influence on decisions was measured by a nine-item index utilizing a 7-point, anchored, Likert scale (Al-Mubarak and Montanari, 1985) that was designed to tap the purchasing agent's inclination to solve current problems by using specific boundary-spanning activities and his/her influence in having subordinates and peers accept these solutions (adapted from Montanari, 1979). Finally, boundary-spanning activities were assessed using Jemison's 25-item scale based on his formulation. This instrument used 5-point, anchored, Likert-type scales (Jemison, 1978, 1979). Means, stan-

Table II. Scale Characteristics, Means, Standard Deviation, Internal Consistency Reliabilities, and Intercorrelation^a

Variable	Range	Number of mean items	Standard deviation	Variable							
				1	2	3	4	5	6	7	8
1. Interdependence											
Substitutability	1-6	12	2.63	.85	(.80)						
Essentiality	1-5	12	2.82	.71	.32 ^e	(.80)					
2. Environmental uncertainty											
Dynamism	1-15	2	7.74	1.20	.02	.15 ^c	(.75)				
Complexity	1-3000	12	189.90	43.50	.24 ^d	.50 ^e	(.76)				
Boundary spanning	1-15	25	8.83	2.30	-.01	.16 ^c	.14 ^c	(.90)			
Face activities	1-5	9	2.01	1.05	-.08	.19 ^c	.03	.17 ^c	(.84)		
Process activities	1-5	7	3.33	.87	.03	.02	.18 ^c	.02	.67 ^e	(.84)	
Relation activities	1-5	9	3.48	1.12	.09	.03	.11 ^b	.16 ^c	.78 ^e		(.76)
4. Influence on decisions	1-12	9	5.70	2.06	.02	.06	-.13	.17 ^c	.20 ^c	-.14 ^b	(.89)

^aOne-tailed test of significance. Numbers in parentheses refer to Cronbach's Alpha. *N* = 133.

^b*p* < .10.

^c*p* < .05.

^d*p* < .01.

^e*p* < .001.

dard deviations, scale characteristics, reliabilities, and zero-order intercorrelations are presented in Table II.

Several analysis techniques were used to investigate the relations postulated in the model. Correlations and path-analytic techniques were used to investigate the implied causal relationships. In addition, moderated regression was used to determine the moderating influence of the IOD variable.

RESULTS

Table II indicates that the correlation between substitutability and dynamism does not reach conventional levels of statistical significance. The significant correlation between substitutability and complexity suggests that subjects perceived their supplier environment as more complex when alternative sources were present. The significant correlation between essentiality and dynamism indicates that the higher the degree of importance of environmental elements to organizational goals, the greater the perceived rate of change in the task environment. As shown in Table II, the correlation between essentiality and complexity is highly significant, which also supports Pfeffer and Salancik's model. The correlation between dynamism and boundary-spanning activity is positive and marginally significant. The reported correlation between complexity and boundary spanning is positive and significant, which provides support for the contention that when the task environment is heterogeneous, more boundary-spanning activities are required.

Path analysis was used to test for possible sequential linkages among interdependence (classified exogenous), environment uncertainty (indigenous), and boundary-spanning activities (indigenous). The results of the path analysis are presented in Fig. 2.

The direct effects of substitutability on dynamism are small and not significant. However, substitutability has a positive impact on complexity. The direct effects are small and significant only at the .10 level. Essentiality has a positive significant impact on dynamism and complexity. The direct effects of essentiality on complexity are substantial and statistically significant at the .001 level. This large impact is a general indication of the importance of essentiality as a predictor of complexity after taking into account all other independent variables contained in the model.

Dynamism has a positive impact on boundary spanning; the direct effects are moderate and are significant at the .10 level. Complexity has a positive impact on boundary spanning; the direct effects are moderate and are significant at the .05 level. However, they explain only a small part ($R^2 = .04$) of the variation in boundary spanning.

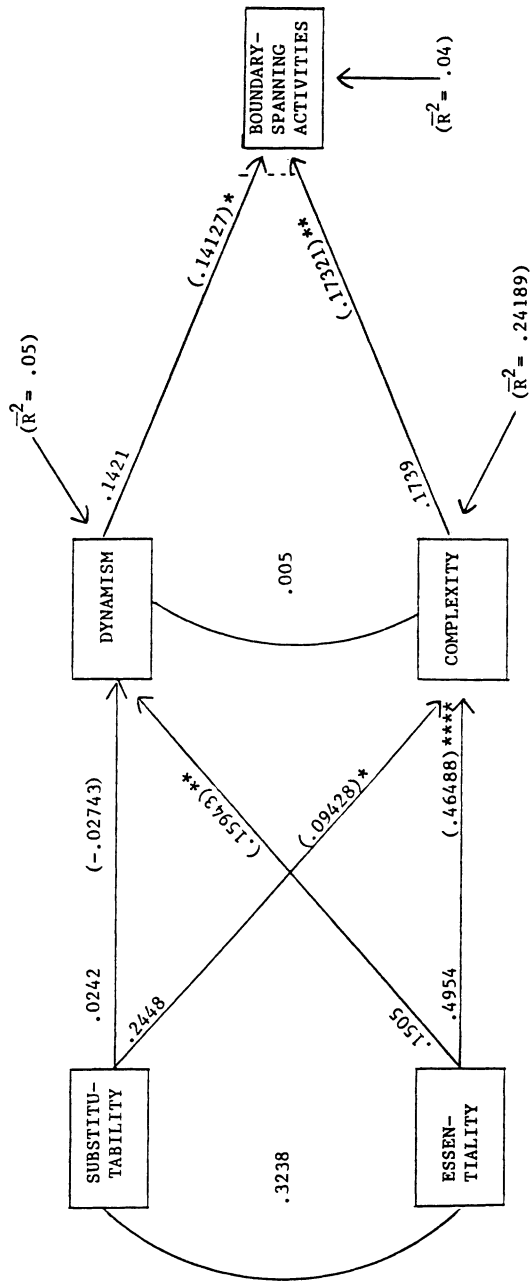


Fig. 2. Path diagram of the hypothesized model. Numbers in parentheses indicate path coefficients (standardized beta). Other numbers are zero-order correlations (* $p < .10$, ** $p < .05$, *** $p < .01$, **** $p < .001$).

To test the accuracy of the fit of the hypothesized model to the observed data, reconstruction of the original zero-order correlations matrix from the estimated parameters was attempted. Unless these reconstructed correlations can be shown to be consistent with the observed correlations, the computed path coefficients may completely misrepresent the causal relationships being investigated and should not be accepted (Billings and Wroten, 1978). The predicted correlations were computed following the procedure outlined by Duncan (1966) and reproduced almost perfectly the observed zero-order correlations which suggests that the proposed model (Fig. 2) included all significant coefficients (Griffin, 1977).

The path analysis results exhibit consistent evidence that substitutability and essentiality directly affect complexity. Also, complexity and dynamism directly affect the relation activity and total boundary spanning. Complexity directly affects face activity, and dynamism directly affects process activity. Furthermore, the variables in the model explain moderate amounts of the variance.

A moderated regression technique was used to test for the hypothesized moderating influence of the IOD variable (Arnold, 1982; Champoux & Peters, 1980). Two equations describe the relationships among the two dimensions of interdependence (substitutability [SUBS] and essentiality [ESSEN]), the two dimensions of environmental uncertainty (dynamism [DYN] and complexity [COMP]), and influence on decisions (IOD). SUBS and ESSEN are the two independent variables. DYN is the dependent variable for the first equation, and COMP is the dependent variable for the second equation. IOD is the moderator variable. The following results were obtained:

$$\begin{aligned}
 1. \text{ DYN} &= .70 - .47 \text{ SUBS} + .08 \text{ ESSEN} + 1.41 \text{ IOD} - .65 \text{ SUBS} * \text{IOD} - .75 \\
 &\quad \text{ESSEN} * \text{IOD} + .25 \text{ SUBS} * \text{ESSEN} * \text{IOD} \\
 &\quad t \quad (7.5) \quad (-1.4) \quad (.21) \quad (.61) \quad (.67) \quad (-.95) \quad (.78) \\
 &\quad b \quad (1.1) \quad (.45) \quad (.41) \quad (.29) \quad (.97) \quad (.79) \quad (.32) \\
 2. \text{ COMP} &= -.06 + .81 \text{ SUBS} + .83 \text{ ESSEN} + 105 \text{ IOD} - 26 \text{ SUBS} * \text{IOD} - \\
 &\quad 30 \text{ ESSEN} * \text{IOD} + 8.5 \text{ SUBS} * \text{ESSEN} * \text{IOD} \\
 &\quad t \quad (-.02) \quad (5.79) \quad (6.56) \quad (1.49) \quad (-.87) \quad (-1.25) \quad (.86) \\
 &\quad b \quad (32.0) \quad (14.0) \quad (12.7) \quad (70.6) \quad (30.0) \quad (24.0) \quad (9.0)
 \end{aligned}$$

The two equations indicate that influence on decisions (IOD) and its interactions with SUBS and ESSEN are not significant. Their *t* values are small and do not approach significant levels.

To test for the moderating effect of IOD in the environmental uncertainty-boundary-spanning activities relationship, a second moderated regression was performed. Three equations describe the relationships among perceived environmental uncertainty (dynamism [DYN] and complexity [COMP]) and boundary-spanning activities (face [FACE], process [PROC], and relation [RELAT]), with influence on decisions [IOD]. DYN and COMP are the two independent variables. FACE, PROC, and RELAT are the de-

pendent variables for the first, second, and third equations, respectively. Following are the three equations and their estimates:

1. $FACE = 7.28 - .70 DYN - .02 COMP - 7.4 IOD + .7 DYN*IOD + .02 COMP*IOD - .02 DYN*COMP*IOD$
 $t \quad (2.2) \quad (-1.7) \quad (-1.6) \quad (-.95) \quad (.74) \quad (.73) \quad (-.46)$
 $b \quad (3.5) \quad (.44) \quad (.01) \quad (7.8) \quad (1.0) \quad (.02) \quad (.04)$
2. $PROC = 8.1 - .61 DYN - .02 COMP - .5 IOD + .74 DYN*IOD + .07 COMP*IOD - .02 DYN*COMP*IOD$
 $t \quad (3.0) \quad (-1.8) \quad (-1.9) \quad (-.98) \quad (.97) \quad (.82) \quad (-.83)$
 $b \quad (2.6) \quad (.34) \quad (.01) \quad (.01) \quad (.76) \quad (.02) \quad (.02)$
3. $RELAT = 10.6 - 1.0 DYN - .02 COMP - 10.0 IOD = 1.36 DYN*IOD + .03 COMP*IOD - .02 COMP*DYN*IOD$
 $t \quad (3.3) \quad (-2.6) \quad (-2.3) \quad (-1.5) \quad (1.51) \quad (1.54) \quad (-1.54)$
 $b \quad (3.1) \quad (.4) \quad (.01) \quad (7.0) \quad (.90) \quad (.02) \quad (.02)$

The three equations indicate that influence on decisions (IOD) and its interactions with DYN and COMP are not significant. Their t values are small and do not approach significant levels.

The absence of a moderating effect for IOD in the previous analyses suggests that another relationship among environmental uncertainty, boundary-spanning activity, and influence on decisions may be operating for this data set. This possibility was reinforced by the fact that IOD was significantly positively correlated with complexity and face-type activity but negatively correlated with process activity (see Table II). To empirically derive the relationship that exists between IOD and complexity, influence on decisions was introduced as a predictor. Multiple regression was performed with substitutability (SUBS), essentially (ESSEN), and influence on decisions (IOD) as independent variables and complexity as the dependent variable:

$$\text{Complexity} = 158 \frac{1}{3} + 15 \text{ SUBS} + 22 \text{ ESSEN} + 18 \text{ IOD}$$

$$t \quad (10.1) \quad (3.2) \quad (4.7) \quad (2.68)$$

$$b \quad (15.7) \quad (4.70) \quad (4.80) \quad (7.02)$$

Influence on decisions is significant at the .01 level when considered a predictor of complexity. Purchasing managers perceived their environments to consist of less substitutable elements of higher importance. If they felt that their choices of spanning activities would be accepted, they perceived the supplier environment as being more complex. This could connote a form of attribution. That is, if the boundary spanners perceive that they have influence, it must be because they cope with a complex environmental element.

Multiple regression was also performed with influence on decisions (IOD), dynamism (DYN), and complexity (COMP) as independent variables and with face activity as a dependent variable:

$$\text{Face} = .96 + .03 \text{ DYN} + .02 \text{ COMP} + .50 \text{ IOD}$$

$$t \quad (1.18) \quad (.42) \quad (1.48) \quad (2.25)$$

$$b \quad (.819) \quad (.078) \quad (.01) \quad (.19)$$

The t value for IOD is large and significant at the .05 level. Other t values are small and do not reach significance. Finally, multiple regression was performed with IOD, dynamism (DYN), and complexity (COMP) as independent variables and with process activity as a dependent variable:

$$\begin{array}{rcccc} \text{Process} & + & 2.38 & + & .13 \text{ DYN} & + & 2 \text{ COMP} & + & .07 \text{ IOD} \\ t & (3.87) & (.42) & (.01) & (-.48) \\ b & (.60) & (.05) & (.01) & (.15) \end{array}$$

The t value for (IOD) and complexity are negative, small, and do not approach the significance level. The t value for dynamism (DYN) is large and significant at the .05 level. Therefore, complexity and influence on decisions do not predict process activity, but dynamism (DYN) is a good predictor of this type of boundary-spanning activity.

DISCUSSION

Results provided only partial support for the theoretical framework presented in Fig. 1. Contrary to Pfeffer and Salancik's (1978) contention, no significant relationship was found between substitutability and dynamism. Significant relationships were found among essentiality, dynamism, and complexity and in the predicted directions. These findings partially support Pfeffer and Salancik's (1978) theory.

The posited relationships between environmental uncertainty and boundary-spanning activity were supported by the data. A significant relationship exists among dynamism, complexity, and boundary spanning. The empirical findings of this research support the theoretical model of Thompson (1962, 1967) and the empirical findings of Cox et al. (1978). Dynamism and complexity appear to affect boundary-spanning activity.

Contrary to the theoretical framework developed here, influence on decisions does not appear to moderate the relationships among dependence, environmental uncertainty, and boundary-spanning activity. One possible reason for the absence of moderating effects is that the study was performed in one industry (electronics). If other industries had been considered, influence on decisions may have emerged as a moderator.

However, several other researchers found significant relationships among environmental uncertainty, boundary-spanning activity, and influence on decisions (Cox et al., 1978; Jemison, 1978; Reiman, 1975). Furthermore, several management theoreticians have incorporated individual influence in the boundary-spanning construct (Strauss, 1962; Thompson, 1967; Pettigrew, 1972; Dubin, 1963; Aldrich & Herker, 1977; Miles, 1980). Boundary spanners monitor the environment and bring information about environmental contingencies to the organization. This information can be a source of in-

fluence for the boundary spanner in the organization. The effective control of dependent and uncertain input environments will probably increase the boundary spanner's influence and, therefore, ability to select and implement his/her chosen activities at the organization's boundary. That is, s/he will be influential enough to interact with the environment as s/he deems appropriate. This can be via face, process, or relation activities. However, these results failed to support this conventional wisdom.

The last three analyses reported here indicate the complex nature of the influence of individual characteristics of boundary spanners on the boundary-spanning process. It appears that the various activities that the boundary spanner performs vary in their susceptibility to individual influence. That is, some activities, i.e., representing and protecting the organization, are engaged in because the boundary spanner perceives that s/he wants to do them and has the power to do so. Results obtained here appear contrary to the notion that such activities are in response to specific environmental demands as Jemison (1979) argued. This could reflect an actual disregard for the dictates of the environment or attribution on the part of the boundary spanner. It may be that boundary spanners in this sample attributed their success in representing or protecting the organization to their efforts even though they may, in actuality, be responding to the environment. For more mundane activities such as gatekeeping, boundary spanners may be willing to admit an external attribution from the environment. The boundary-spanning specific nature of these results suggests that additional research is necessary to determine precisely what factors act as antecedents to specific boundary-spanning activities.

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